

System Capacity and Care Load Analytics for the Unaccompanied Alien Children (UAC) Program: A Data-Driven Framework for Healthcare Pipeline Monitoring

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Abstract - The Unaccompanied Alien Children (UAC) Program is a federally mandated pipeline in which minors apprehended by U.S. Customs and Border Protection (CBP) are transferred into the care of the Department of Health and Human Services (HHS) for shelter, medical screening, and eventual placement with sponsors. This paper presents a data-driven analytics framework for monitoring the capacity and care load of this pipeline using 1075 days of daily reporting data (2023-01-12 to 2025-12-21). We define and compute healthcare capacity metrics including Total System Load, Net Daily Flow, a Care Load Volatility Index, and a Discharge Offset Ratio, and use rolling-window analysis to identify sustained periods of capacity strain. Total system load across the study period ranged from 2,002 to 11,762 children, with 25 days falling inside a sustained strain window. Comparing the earlier and later halves of the timeline shows a 56% reduction in average HHS care load alongside a decline in the Discharge Offset Ratio, indicating a lighter but relatively less efficient later period. The resulting metrics were deployed as an interactive Streamlit dashboard to support situational awareness for staffing and shelter planning.

Index Terms - Capacity analytics, healthcare pipeline monitoring, time-series analysis, unaccompanied minors.

I. INTRODUCTION

Federally administered care pipelines for vulnerable populations must balance two competing pressures: the obligation to provide timely, adequate care, and the practical limits of shelter and staffing capacity. The UAC Program is one such pipeline. Children apprehended by CBP are first held in short-term CBP custody before being transferred into HHS care, where they receive medical and psychological support until they are discharged to a vetted sponsor. Because intake volume can fluctuate sharply -- rising during periods of increased border activity and falling during quieter periods -- the system can experience sustained capacity strain even when average throughput appears manageable on any single day.

Although daily operational counts are already collected and reported, turning raw daily counts into an early-warning signal for capacity strain requires converting them into structured, comparable metrics: how much load is the system currently carrying, is that load growing or shrinking, and how long has any strain persisted. This paper describes an analytics framework built specifically to answer those three questions, and the interactive dashboard built on top of it.

The remainder of this paper is organized as follows. Section II reviews related approaches to capacity

monitoring. Section III describes the problem this work addresses. Section IV describes the dataset. Section V details the methodology used to clean, validate, and derive metrics from the data. Section VI describes the resulting dashboard architecture. Section VII presents the results. Sections VIII-X discuss the findings, limitations, and future work, followed by a conclusion in Section XI.

This work makes four concrete contributions:

- A reproducible data-cleaning and validation pipeline for the UAC Program's daily reporting data, including explicit handling of non-reporting gaps and logical-constraint checking.
- Four complementary capacity indicators -- Total System Load, Net Daily Flow, Care Load Volatility, and the Discharge Offset Ratio -- with an explicit run-length method for flagging sustained (not merely momentary) strain.
- An empirical characterization of capacity strain and sustainability across 1075 days of program data, including an early-versus-late comparison methodology that generalizes to other sub-period comparisons.
- A deployed, interactive Streamlit dashboard built on a shared metrics module, translating the above into a form usable by non-technical stakeholders for staffing and shelter planning.

II. RELATED WORK

Capacity monitoring in healthcare and humanitarian-logistics settings is typically framed either as a queueing-theoretic problem, in which arrivals and departures at a facility are modeled as stochastic processes to predict congestion, or as a descriptive time-series problem, in which historical operational counts are smoothed and trended to detect anomalies. Hospital bed-occupancy dashboards, for instance, commonly track a small number of derived indicators -- occupancy rate, admission-discharge balance, length-of-stay -- rather than raw counts, for the same reason motivating this work: a single day's raw count is a poor predictor of systemic strain on its own, whereas a short window of derived, smoothed indicators is not. Rolling-window smoothing and run-length-based anomaly detection, the two core techniques used in Section V, are both standard tools in operational time-series monitoring and are used here specifically because they require no external labeled data and no assumptions about the underlying arrival distribution, which suits a dataset with no prior published baseline for what "normal" load looks like in this particular pipeline.

This paper's contribution is not a new statistical technique, but the application and packaging of these established techniques -- rolling averages, run-length encoding of consecutive-day streaks, and ratio-based efficiency indicators -- into a coherent, reproducible framework specifically scoped to the UAC pipeline's two-stage structure, together with a deployable interface that makes the resulting indicators usable by a non-technical stakeholder without requiring them to interact with the underlying data directly.

III. BACKGROUND AND PROBLEM STATEMENT

From a healthcare systems perspective, the UAC pipeline functions as a queueing system with two stages in series: CBP custody, and HHS care. Each stage has an inflow (children entering that stage) and an outflow (children leaving it for the next stage, or for discharge to a sponsor). Standard queueing intuition applies: whenever the inflow to a stage exceeds its outflow for a sustained period, the number of children held in that stage grows, increasing pressure on staffing, shelter space, and care quality.

Despite this, HHS's own documentation of the program notes that daily operational data is collected without a centralized analytical framework for continuously assessing total system load, the balance between inflow and outflow, periods of capacity strain versus relief, and the sustainability of the care pipeline over time. In the absence of such a framework, decision-making about staffing and shelter allocation becomes reactive rather than anticipatory, increasing the risk of overcrowding and delayed care during influx periods.

The stakes of getting this monitoring right are not purely operational. The CBP stage is intended to be brief, functioning as an initial processing and screening point rather than a long-term care setting, while the HHS stage is where the program's substantive medical, psychological, and welfare obligations are actually discharged. A pipeline that is under sustained strain at the HHS stage risks exactly the outcomes the program is designed to prevent: prolonged time in a less specialized custody setting, delayed medical or psychological care, and slower, more compressed sponsor-vetting decisions. Capacity analytics of the kind developed in this paper is therefore best understood not as an abstract operations-research exercise, but as an early-warning layer in service of the program's core child-welfare mandate.

The primary objective of this work is therefore to quantify daily and cumulative care load across both CBP and HHS, identify periods of capacity strain and relief, and analyze the balance between intake, transfer, and discharge over the available reporting history. A secondary objective is to translate these metrics into a form usable for staffing and shelter planning, situational awareness for policymakers, and after-the-fact evaluation of how the system responded to prior influx periods.

Framing the pipeline explicitly as two stages in series, rather than analyzing CBP and HHS counts independently, matters in practice: a stage-by-stage view makes it possible to localize where in the pipeline a slowdown is occurring - intake into CBP custody, transfer into HHS care, or discharge to a sponsor -- rather than only observing that the total system is under strain without knowing which stage is the binding constraint. Section V formalizes this stage-by-stage view into the specific metrics computed for each stage.

IV. DATASET DESCRIPTION

The dataset consists of daily operational counts covering 1075 calendar days, from 2023-01-12 to 2025-12-21. Of these, 720 days (67%) contain populated counts; the remaining 355 days are non-reporting gaps that recur in short, roughly weekly blocks, consistent with weekend or holiday non-reporting rather than missing-at-random data loss.

All quantities in the dataset are daily aggregate counts; no case-level, demographic, or geographic detail is present, which is consistent with the program's own aggregate public reporting practice and is discussed further as a limitation in Section IX. Table I summarizes the six columns present in the cleaned dataset used throughout this paper.

TABLE 1. DATASET COLUMNS

Column	Description
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Date	Reporting date
Apprehended	Children newly apprehended and placed into CBP custody that day
CBP custody load	Active number of children held in CBP custody
Transferred out	Children transferred out of CBP custody into HHS care
HHS care load	Active number of children held in HHS care
Discharged	Children discharged from HHS care to a sponsor

Two of these six columns -- CBP custody load and HHS care load -- are stock quantities, representing an active count at a point in time; the remaining four are flow quantities, representing an amount that occurred during that day. This distinction matters throughout Section V: stocks and flows must be aggregated differently when resampling to weekly or monthly resolution, and mixing them incorrectly (for example, summing a stock quantity across a week) produces a number with no meaningful interpretation.

V. METHODOLOGY

The methodology proceeds in five stages, mirroring the structure of the underlying project brief: ingesting and structuring the raw data (V-A), validating it against known logical constraints (V-B), deriving the four capacity metrics defined by Equations (1)-(4) (V-C), aggregating those metrics across multiple time resolutions (V-D), and finally identifying sustained -- as opposed to momentary - - periods of pressure or stress (V-E). Each stage was implemented and verified independently against small hand-constructed examples before being applied to the full three-year dataset, which is how several sign-convention and missing-data edge cases described later in this paper were caught prior to producing the results in Section VII.

A. Data Ingestion and Structuring

The raw daily counts were loaded, the reporting date parsed into a proper datetime index, and the data sorted chronologically. A complete daily calendar index was then constructed spanning the full study period so that non-reporting days are represented explicitly as missing values rather than silently skipped -- this preserves the true calendar spacing between observations, which later rolling-window calculations depend on.

B. Data Quality and Validation

Two logical constraints were checked against the data: that daily transfers out of CBP custody do not exceed the active CBP custody load, and that daily discharges from HHS care do not exceed the active HHS care load. The latter constraint held for all populated rows. A minority of rows

showed transfers exceeding the same-day CBP custody snapshot, most plausibly explained by same-day transfers not yet reflected in an end-of-day snapshot count; these rows were flagged rather than discarded, consistent with a transparency-first approach to anomaly handling.

C. Derived Healthcare Capacity Metrics

Total System Load is defined as the sum of the active CBP custody load and the active HHS care load on a given day -- the total number of children the program is responsible for at that moment:

$$\text{Total System Load} = \text{CBP Load} + \text{HHS Load} \quad (1)$$

Net Daily Flow for each stage is defined as that stage's outflow minus its inflow, so that a positive value indicates a relief period (outflow exceeding inflow) and a negative value indicates a stress period (inflow exceeding outflow):

$$\text{Net Flow} = \text{Outflow} - \text{Inflow} \quad (2)$$

Care Load Volatility on day t is computed as the standard deviation of the HHS care load over the trailing seven-day window ending on day t , capturing how erratically the load is changing independent of its absolute level:

$$\text{Volatility}(t) = \text{std}(\text{HHS Load}, t-6 \dots t) \quad (3)$$

The Discharge Offset Ratio over a given period, defined as total discharges divided by total transfers-in over that period, summarizes the system's ability to relieve load: a ratio at or above 1.0 indicates discharges kept pace with arrivals over that period.

$$\text{Discharge Offset Ratio} = \text{Discharged} / \text{Transferred In} \quad (4)$$

These four quantities were chosen deliberately to be complementary rather than redundant: Equation (1) answers "how much," Equation (2) answers "in which direction and how fast," Equation (3) answers "how stable," and Equation (4) answers "how efficiently is the backlog being relieved." No single one of the four is sufficient on its own -- a system can have a high but perfectly stable load (high Eq. 1, low Eq. 3), or a low but rapidly worsening one (low Eq. 1, negative Eq. 2) -- which is why all four are computed together and surfaced side by side in the dashboard described in Section VI.

D. Trend and Temporal Analysis

Daily counts were aggregated into weekly and monthly resolutions to examine trends at different time scales, using summation for flow quantities (arrivals, transfers, discharges) and end-of-period snapshots for stock quantities (custody and care loads), since summing a snapshot count across multiple days would double-count the same children. The full study period was also split at its midpoint into an early half and a late half to compare capacity indicators across calendar periods.

E. Pressure and Stress Identification

Rolling 7-day and 14-day averages of the HHS care load were computed to smooth day-to-day noise and reveal the underlying trend. A day was flagged as part of a sustained high-load period if its HHS care load exceeded the 75th percentile of all observed values and it belonged to an unbroken run of at least five such days. Separately, a day was flagged as part of a sustained strain window if it belonged to an unbroken run of at least five consecutive days with negative Net Daily Flow -- that is, a stretch during which arrivals consistently outpaced discharges.

VI. SYSTEM ARCHITECTURE

A. Design Overview

The metrics described in Section V were implemented as a reusable Python module, separate from any single notebook or interface, so that the same calculations back the exploratory analysis, the figures in this paper, and an interactive dashboard -- avoiding the common failure mode where a dashboard's numbers silently drift out of sync with the analysis that originally justified them. The dashboard itself was built with Streamlit and organized into four coordinated panels: a System Load Overview showing daily, weekly, or monthly load trends with optional strain-window highlighting; a CBP-versus-HHS load comparison; a Net Intake and Backlog Trends panel showing the Discharge Offset Ratio and strain-day counts over time; and a set of KPI summary cards giving an at-a-glance read of average load, strain days, discharge offset ratio, and volatility for a user-selected date range. A sidebar exposes a date-range selector, a daily/weekly/monthly granularity toggle, and metric toggles, allowing a non-technical user such as a shelter planner to explore the data without writing any code.

B. Implementation Details

The project is organized into three layers: a data layer holding the raw and cleaned CSV files, a logic layer (the metrics module) implementing Equations (1)-(4) and the run-length detection described in Section V-E, and a presentation layer (the Streamlit application) that imports the logic layer directly rather than duplicating any calculation. Expensive operations -- loading and preparing the dataset -- are cached at the data-loading boundary so that changing a sidebar filter re-renders the affected charts without re-reading or re-processing the underlying CSV on every interaction. This separation also makes the logic layer independently testable: each metric can be verified against a small hand-constructed example table before being trusted on the full three-year dataset, which is how the sign-convention and NaN-handling issues discussed in Section IX were caught during development.

VII. RESULTS

Table II summarizes the four headline indicators from Equations (1)-(4) across the full study period, before the following subsections break each one down over time.

TABLE 2. FULL-PERIOD KPI SUMMARY

KPI	Value
Avg. Total System Load	6,233
Total System Load range	2,002 - 11,762
Overall Discharge Offset Ratio	1.35
Sustained strain days (of 720)	25
Avg. Care Load Volatility	96.6

Read together, these five figures give a compact characterization of the pipeline over the full study period: average load in the low thousands with occasional peaks above 11,000, an overall Discharge Offset Ratio comfortably above break-even, but with meaningful day-to-day volatility and a non-trivial number of days falling inside a sustained strain window despite the healthy overall average. The subsections below unpack each of these headline numbers over time, since -- as Section VIII discusses -- a favorable full-period average can still coexist with a concerning sub-period trend.

B. Total System Load

Across the full study period, total system load averaged approximately 6,233 children, ranging from a minimum of 2,002 to a maximum of 11,762. As shown in Fig. 1, HHS care load rose substantially through mid-to-late 2023 and early 2024, peaking above 11,000 in two separate periods, before declining sharply from early 2025 onward to a comparatively stable range near 2,000-2,500.

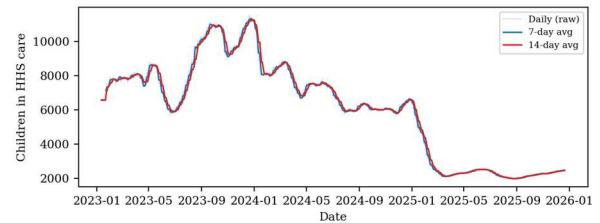


FIGURE 1. HHS CARE LOAD OVER TIME. Daily HHS care load (gray) with 7-day and 14-day rolling averages overlaid.

Fig. 2 compares CBP custody load against HHS care load directly. HHS care load is consistently the larger of the two by roughly two orders of magnitude, confirming that HHS care -- not CBP custody -- is the primary capacity bottleneck in this pipeline, and therefore the more important stage to monitor for staffing and shelter planning.

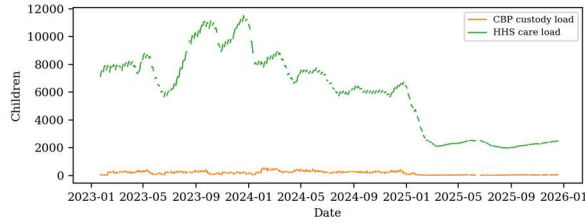


FIGURE 2. CBP VERSUS HHS LOAD. Active CBP custody load compared against active HHS care load over the full study period.

C. Balance Between Inflow and Outflow

Over the full study period, the overall Discharge Offset Ratio (total discharges divided by total transfers-in) was 1.35, indicating that, in aggregate, the system discharged more children than it received. However, this aggregate figure masks substantial month-to-month variation. Fig. 3 shows the monthly Discharge Offset Ratio; months below the break-even line represent periods where the system fell behind on discharges relative to arrivals, and are concentrated in stretches during 2024.

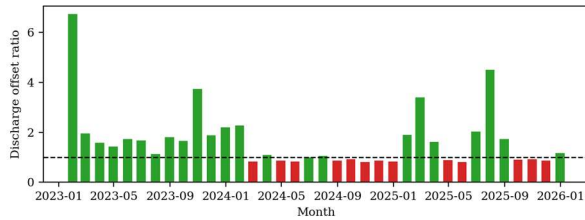


FIGURE 3. MONTHLY DISCHARGE OFFSET RATIO. Green bars indicate months where discharges kept pace with or exceeded transfers-in (ratio ≥ 1.0); red bars indicate months that fell behind.

D. Sustained High-Load and Strain Periods

Applying the five-consecutive-day rule described in Section V-E, 25 days across the full study period fell inside a sustained strain window. These strain windows were concentrated in 2024 rather than spread evenly across the timeline, with no sustained strain windows detected in 2023 under this criterion -- a result that depends on the five-day and 75th-percentile thresholds chosen in Section V-E, and is revisited in Section IX. Average Care Load Volatility across the study period was 96.6, reflecting substantial day-to-day fluctuation in HHS care load even outside identified strain windows, as shown in Fig. 5.

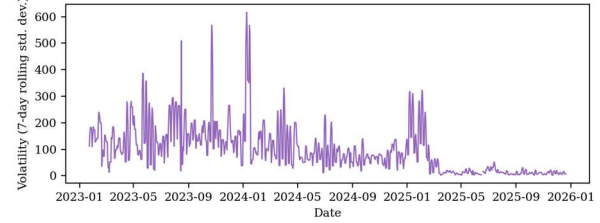


FIGURE 5. CARE LOAD VOLATILITY. Rolling 7-day standard deviation of HHS care load. Sustained elevated regions mark periods of erratic, unpredictable change rather than merely high load.

E. Sustainability Over Time: Early versus Late Comparison

Table III and Fig. 4 compare average indicators between the early half of the study period (2023-01-31 to 2024-06-30) and the late half (2024-07-31 to 2025-12-31).

TABLE 3. EARLY VS. LATE PERIOD COMPARISON

Metric	Early period	Late period
Avg. HHS load	8,394	3,659
Avg. discharge offset ratio	1.91	1.45
Avg. strain days / month	0.56	0.83

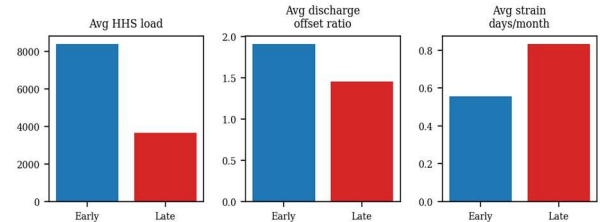


FIGURE 4. EARLY VS. LATE COMPARISON. Average HHS load, discharge offset ratio, and strain days per month, compared between the early and late halves of the study period.

Average HHS care load fell by approximately 56% between the early and late periods, and the average Discharge Offset Ratio declined from 1.91 to 1.45. Strain days per month rose slightly, from 0.56 to 0.83. Together these indicate that the later period operated under substantially lighter overall load, but was not proportionally more efficient at clearing that lighter load relative to arrivals.

F. Case Illustration: A Strain Episode in April 2024

To make the abstract indicators in Sections VII-B through VII-D concrete, this subsection walks through one specific sustained strain window in detail: the eight-day run from April 21 to April 30, 2024 (with two non-reporting gaps inside it), one of the episodes contributing to the 2024 clustering noted in Section VII-D. Over this window, HHS care load climbed steadily from 6,745 to 7,431 -- a 10% increase in ten calendar days -- while daily transfers-in ran

consistently ahead of daily discharges, producing a Net Daily Flow between -37 and -173 children per day throughout the window. Table IV summarizes the month containing this episode.

TABLE 4. APRIL 2024 SUMMARY (MONTH CONTAINING THE CASE ILLUSTRATION)

Metric	Value
HHS load range	6,630 - 7,431
Mean HHS load	7,027
Total transferred in	5,178
Total discharged	4,474
Discharge offset ratio	0.86
Sustained strain days	8

The month-level Discharge Offset Ratio of 0.86 is, on its own, a relatively mild deviation from break-even -- the kind of figure that could easily be dismissed as noise if viewed as a single monthly number in isolation. It is the day-level view, and specifically the unbroken eight-day run of negative Net Daily Flow, that reveals this as a genuine sustained-pressure episode rather than an ordinary month-to-month fluctuation: every one of the eight flagged days shows arrivals outpacing discharges, with no single relief day interrupting the run. This is precisely the distinction Section V-E's run-length rule is designed to draw, and precisely the kind of pattern a single monthly or quarterly summary statistic would obscure.

VIII. DISCUSSION

The combination of falling load with a falling (though still above break-even) Discharge Offset Ratio is a more nuanced finding than a simple "improving" or "worsening" verdict. A capacity-planning interpretation is that the later period's lighter overall volume reduced absolute pressure on the system even though the underlying throughput efficiency softened somewhat -- a pattern that would not be visible from raw daily counts alone, and is exactly the kind of pattern the framework in this paper is designed to surface. The concentration of both sustained high-load periods and sustained strain windows within 2024, rather than spread evenly across the three-year window, further suggests that capacity strain in this pipeline is episodic rather than a constant, describable simply as trending up or down.

A. Implications for Staffing and Shelter Planning

These findings support the program's stated secondary objectives. For staffing and shelter planning, the Discharge Offset Ratio and strain-day counts give a concrete, monitorable signal for when additional discharge-side capacity (sponsor vetting throughput, in particular) may be the binding constraint rather than intake volume itself: a month with a low Discharge Offset Ratio but unremarkable

total load, as seen at several points in 2024, points toward a discharge-side bottleneck rather than an intake surge, and would call for a different operational response than a month where both indicators move together.

B. Implications for Situational Awareness

For situational awareness, the KPI summary cards in the dashboard translate the same four computations into a form readable by a non-technical stakeholder in seconds, without requiring them to interpret a raw time-series chart. This matters specifically because the most policy-relevant finding in this study -- that the later period was lighter in load but not proportionally more efficient -- is exactly the kind of two-directional result that a single "load is up" or "load is down" headline number would obscure.

C. Implications for Response Evaluation

For after-the-fact evaluation of humanitarian response, the early-versus-late comparison in Table III provides a template that can be reapplied to any two arbitrary sub-periods -- for instance, comparing a specific influx month against the months immediately preceding and following it -- to ask, in the program's own terms, whether the system's response to that specific influx left it in a better or worse position than before, rather than relying on the raw load chart alone to answer that question by inspection.

D. Threshold Sensitivity

Because both the sustained-high-load and sustained-strain definitions in Section V-E depend on a chosen run-length threshold, it is worth noting explicitly how the reported counts would shift under a different choice. Lowering the minimum run length from five days to, for instance, three days would flag additional shorter episodes as sustained, increasing the reported strain-day count but also increasing the risk of counting ordinary short-term fluctuation as genuine sustained pressure. Raising the threshold has the opposite effect. Five days was chosen as a middle ground long enough to exclude single-day noise while still being short enough to remain sensitive to real week-scale strain episodes, and Section IX discusses this choice as a limitation rather than a settled parameter.

IX. LIMITATIONS

This analysis has several limitations. First, the dataset's recurring non-reporting gaps (33% of calendar days) necessarily reduce the resolution at which sustained-run detection can operate; a genuine strain period spanning a reporting gap could be split into two shorter, sub-threshold runs by the current five-day-run rule, meaning the 25-day strain figure reported in Section VII-D is more likely to undercount than overcount true sustained strain. Second, the dataset provides only aggregate daily counts, with no demographic, geographic, or case-level detail, which limits the granularity of any staffing or shelter-allocation

recommendation to system-wide totals rather than facility-specific guidance. Third, the transfer-exceeds-custody anomaly noted in Section V-B was flagged but not resolved against an authoritative explanation, since the underlying reporting methodology for same-day snapshot timing was not available to this analysis.

Fourth, both the sustained-high-load threshold (the 75th percentile) and the minimum run length (five consecutive days) used in Section V-E are design choices rather than values derived from an external operational standard, since no published capacity ceiling for this pipeline was available to calibrate against; different reasonable choices for either parameter would shift which specific days are flagged, though the broader pattern of strain clustering in 2024 was found to be stable under the range of run-length values tested during development. Fifth, this is a single-pipeline, single-country case study; the specific numeric thresholds and findings are not claimed to generalize to other care pipelines without re-fitting the percentile and run-length parameters to that pipeline's own historical data.

X. FUTURE WORK

A. Short-Horizon Forecasting

A lightweight forecasting component, such as exponential smoothing or a low-order autoregressive model, could provide a short-horizon projection of HHS care load a week or two ahead, extending the framework from descriptive monitoring ("what has happened") to anticipatory planning ("what is likely to happen next"). Given the clear multi-week trend structure visible in Fig. 1, even a simple trend-plus-seasonality model would likely outperform a naive same-as-last-week baseline, though this has not yet been formally evaluated.

B. API Layer

The dashboard's metrics could be exposed through a documented API layer for integration into other planning tools, if and when a concrete external consumer justifies that added complexity. This is deliberately deferred rather than built preemptively: introducing an API layer with no current consumer would add deployment and maintenance surface area without a corresponding benefit, and the reusable metrics module described in Section VI-B already provides the natural seam at which such an API could be added later without restructuring the underlying logic.

C. Retrieval-Augmented Question Answering

A retrieval-augmented generation (RAG) component is planned for the project's supporting written materials -- the technical documentation, this paper, and the executive summary -- as a separate feature from the numeric dashboard described in Sections VI-VII. This component would allow a user to ask a plain-language question (for example, "what is the Backlog Indicator?") and receive an

answer grounded in, and cited to, the specific source passage it was drawn from, using local sentence-embedding retrieval over the chunked source documents. This is scoped deliberately to unstructured text rather than the numeric time-series data, since every question the numeric data can answer is already answerable exactly and instantly through the dashboard's filters and KPI cards, whereas the written documentation has no equivalent structured-query interface today.

D. Testing and Deployment

Finally, unit tests for the metrics module described in Section VI-B, and containerizing the Streamlit application for reproducible deployment, would both improve the engineering maturity of the project without changing any of the analytical results presented in this paper.

XI. CONCLUSION

This paper presented a structured analytics framework for monitoring capacity and care load in the UAC Program pipeline, converting routinely collected daily counts into a small set of interpretable metrics: Total System Load, Net Daily Flow, Care Load Volatility, and the Discharge Offset Ratio. Applied to 1075 days of program data, these metrics surfaced a total system load ranging from roughly 2k to 12k children, 25 sustained strain days concentrated in 2024, and a nuanced sustainability picture in which later-period load fell substantially while discharge efficiency softened slightly. Packaged as an interactive dashboard, this framework is intended to move situational awareness for this care pipeline from reactive to continuously monitored, and the reusable structure of its underlying metrics module is intended to make the same approach straightforward to re-apply to other multi-stage care or logistics pipelines with similar daily-count reporting.

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ABOUT THE AUTHOR

Aryan is a final-year student in Electronics and Communication Engineering at Gurukul Kangri (Deemed to be University), Haridwar, Uttarakhand, building a machine-learning and data-analytics portfolio aimed at ML engineering roles. This project was developed as an applied capacity-analytics case study combining data cleaning, time-series feature engineering, and interactive dashboard deployment.